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Teachers' artificial intelligence (AI) literacy: an exploratory study

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Abstract

This study explores variables associated with teachers' Artificial Intelligence (AI) literacy, a key competency for effective and responsible AI integration in education. A total of 270 teachers completed an online survey including measures of AI literacy, AI acceptance, computational thinking, AI anxiety, and digital divide. Results revealed that all AI acceptance variables were positively associated with AI literacy, with hedonic motivation and willingness to use AI emerging as the strongest predictors. Computational thinking, AI anxiety, and digital divide also showed significant associations with AI literacy. The findings highlight the central role of teachers' attitudes and motivational variables over technical and demographic variables. The study contributes to the understanding of how teachers engage with AI technologies and provides practical implications for designing professional development programs that enhance AI literacy and reduce barriers to AI adoption in educational contexts.

Keywords: Teachers, Artificial intelligence (AI), AI Literacy, AI acceptance, Computational thinking, AI anxiety, Digital divide

Introduction

Since the beginning of 2023, the use of Generative Artificial Intelligence (GenAI) chatbots, such as ChatGPT, has increased significantly. This development underscores the imperative necessity for Artificial Intelligence (AI) literacy as an essential competency in both professional and educational settings (PwC, 2025).

AI literacy has become an essential skill for teachers, enabling them to effectively integrate AI technologies into their teaching practice while maintaining awareness of ethical considerations. Teachers with higher AI literacy levels are better equipped to harness AI's potential for personalized learning and data-driven instruction while also being more capable of addressing important concerns around bias, privacy, and transparency (Zhao et al., 2022). As AI continues to transform educational environments, developing teachers' AI literacy has emerged as a fundamental prerequisite for ensuring equitable and responsible AI integration in education and preparing their students for an AI-driven future. This makes it essential for teacher education programs to incorporate systematic AI literacy training. However, research in this field is still in its early stages (Hur, 2025).

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This study aims to address the critical gap in understanding AI literacy among teachers at a time when AI technologies are rapidly transforming the educational landscape. The study will provide a comprehensive assessment of AI literacy levels among Israeli teachers, establishing a baseline for future research and educational policy development while examining the relationships between AI literacy and variables such as AI acceptance, computational thinking, AI anxiety, and the digital divide to develop a more nuanced understanding of the factors associated with teachers' engagement with AI technologies.

The findings will guide stakeholders aiming to enhance teachers' AI literacy, professional development programs, and teachers' self-planning efforts to advance their AI literacy by tackling specific barriers and challenges they encounter when adopting AI technologies. They will also contribute to the theoretical framework for understanding AI literacy in educational contexts and potentially serve as a model for similar studies in other countries.

By providing educational leaders and policymakers with the necessary knowledge to make informed decisions regarding professional development and curriculum integration, this study seeks to deepen the understanding of teachers' AI competencies and offer critical insights for designing targeted training programs, promoting inclusive learning environments, and bridging knowledge gaps, ultimately equipping educators with the skills to effectively integrate AI tools into their teaching to address potential challenges, support diverse student needs, enhance educational technology, and prepare students for a future shaped by AI.

This research examines the level of AI literacy among teachers in Israel and explores a range of variables associated with AI literacy: (1) AI acceptance, (2) computational thinking, (3) AI anxiety, and (4) the digital divide in AI. These variables have been selected based on their established relationships with technologies previously explored in the research literature. By scrutinizing these connections within the AI domain, this research aspires to enhance existing insights while addressing the distinct challenges and opportunities presented by AI.

Literature review

AI Literacy

The concept of "literacy" has undergone significant evolution over time. It now encompasses not only reading and writing competencies but also essential technological skills that are crucial in contemporary society. Additionally, it includes capabilities such as problem-solving, collaboration, and the ability to effectively present information (Mackey & Jacobson, 2011). In the digitally driven twenty-first century, several types of literacy have been recognized, including information, communication, and digital literacies (Naji et al., 2019; Spires et al., 2012).

AI literacy builds upon and expands previous forms of literacy such as computer, digital, media, information, and technological literacy. These foundational literacies provide a framework for developing AI literacy, which encompasses definitions, theoretical exploration, measurement techniques, and practical applications (Ng et al., 2023).

AI literacy is an evolving field driven by the need for effective integration of AI in education and the workforce (Ding et al., 2024a, 2024b). It includes a broad array of skills and insights that extend beyond mere technical knowledge (Cox, 2024). This comprehensive

concept encompasses understanding AI's potential and limitations, reflecting on its ethical dimensions, and proficient collaboration with AI systems. It necessitates the ability to evaluate, produce, and engage ethically with AI technologies (Celik, 2023; Chiu et al., 2024; Cox, 2024; Long & Magerko, 2020; Ng et al., 2021).

According to a UNESCO report, AI literacy is essential for all citizens across various educational stages, aiming to prepare them for a world increasingly shaped by AI. While GenAI is emerging as a critical skill that can open new opportunities, it requires critical thinking and careful application to avoid negative consequences (UNESCO, 2024).

Several thematic frameworks define the core components of AI literacy. A widely recognized model proposed by Ng et al. (2021) categorizes AI literacy into four key dimensions: (1) Knowing and Understanding AI—grasping AI concepts, including definitions, capabilities, limitations, and future potential; (2) Using and Applying AI—developing practical skills to interact with AI applications for enhancing tasks and achieving goals; (3) Evaluating and Creating AI—distinguishing AI-driven systems from non-AI systems and recognizing the differences between human and AI interactions; and (4) AI Ethics—critically assessing the ethical and societal implications of AI, ensuring responsible and informed decision-making in its application.

This four-dimensional model of AI literacy has been widely adopted and further operationalized in numerous empirical studies. Subsequent research has translated Ng et al.'s (2021) conceptual framework into validated measurement tools that capture these components in diverse educational and professional contexts. For example, Wang et al. (2022) developed the AI Literacy Scale, a psychometric instrument that measures knowledge, application, evaluation, and ethical understanding. As highlighted by Lintner et al. (2024), despite methodological and contextual diversity, most validated instruments converge around these four dimensions: knowledge, application, evaluation, and ethics, indicating an emerging consensus on the multidimensional nature of AI literacy. Collectively, these developments illustrate the field's evolution from conceptual exploration toward empirically grounded, psychometrically sound measurement frameworks (Lintner et al., 2024; UNESCO, 2024; Wang et al., 2022).

Teachers' AI Literacy

AI literacy has emerged as a crucial competency for educators in the digital age, encompassing a broad set of knowledge and skills necessary for understanding, utilizing, and critically evaluating AI technologies within educational contexts (Wijaya et al., 2024). This competency is essential as AI increasingly permeates daily life and educational environments, requiring teachers to effectively navigate and implement AI-enhanced tools (Zhao et al., 2022). Without sufficient AI literacy, educators may struggle to harness AI's potential for personalized learning, data-driven decision-making, and classroom automation, thereby limiting its transformative impact on pedagogy (Younis, 2024). Furthermore, AI literacy extends beyond mere technical knowledge, necessitating critical thinking skills to assess AI-driven recommendations and ethical considerations such as bias, transparency, and privacy (Ding, et al., 2024a). Research highlights that educators with higher AI literacy demonstrate greater confidence in integrating AI tools into their teaching, fostering more adaptive and student-centered learning environments (Hur, 2025). Given the growing prevalence of AI in education, scholars emphasize the need

for systematic AI literacy training in teacher education programs to ensure that educators can effectively leverage AI while addressing its broader societal and ethical implications (Su, 2024). Consequently, promoting AI literacy among teachers is not merely an enhancement of digital competence but a fundamental prerequisite for equitable and responsible AI integration in education.

AI literacy among teachers is increasingly recognized as a multidimensional competency that encompasses a fundamental understanding of AI concepts, the ability to apply AI tools in educational contexts, the capacity to critically evaluate AI technologies and an awareness of the ethical considerations surrounding AI integration in education (Wijaya et al., 2024; Younis, 2024). Research indicates that AI literacy plays a crucial role in enhancing instructional methodologies, fostering student engagement, and ensuring the responsible use of AI-driven educational tools (Ding, et al., 2024a; Hur, 2025). Teachers with higher AI literacy demonstrate a greater ability to critically assess AI applications, integrate AI within pedagogically sound frameworks, and collaborate effectively with AI-based systems (Su, 2024; Zhao et al., 2022).

Scholars emphasize the necessity of comprehensive AI literacy training in teacher education programs to equip educators with the skills needed to navigate and utilize AI effectively while critically assessing its broader implications for education and society (Ding et al., 2024b; Younis, 2024).

Research on teachers' AI literacy remains in its infancy, showing substantial cross-national variation in focus, measurement approaches, and reported literacy levels. In the United States, teachers demonstrated moderate AI literacy and frequent misconceptions about fundamental concepts (Ding et al., 2024a, 2024b). In China, teachers reported moderate-to-high literacy levels, with the ability to apply AI in classroom contexts emerging as a strong predictor of both conceptual understanding and intentions for future AI learning (Du et al., 2024; Zhao et al., 2022). In Turkey, pre-service teachers expressed positive attitudes toward AI tools but reported limited content knowledge and insufficient formal training (Karaduman, 2025). Similarly, in Hong Kong, early childhood teachers displayed openness toward tools such as ChatGPT while expressing low self-efficacy and uncertainty regarding their professional competence (Yim & Wegerif, 2024). Overall, these studies suggest that teachers' AI literacy is shaped by cultural, institutional, and perceptual factors, underscoring the need for context-sensitive assessment tools and tailored professional development initiatives.

Thus, AI literacy is not merely a technical proficiency but a critical pedagogical and ethical competency that underpins the responsible and effective integration of AI in education.

Variables associated with AI literacy among teachers

Previous studies explore the associations between AI literacy and psychological contracts. AI literacy significantly correlates with factors like emotional engagement, self-regulated learning, autonomy, and competence among pre-service teachers, with emotional engagement mediating the connection between basic psychological needs, autonomy and competence, and AI literacy (Yang et al., 2024). Teachers engaged in AI coursework demonstrate stronger AI literacy skills. Self-regulated learning moderates the impact of emotional engagement on AI literacy, indicating that educators who

manage their learning strategies benefit more from AI instruction (Ding et al., 2024b; Yang et al., 2024). Beyond engagement and learning strategies, research also suggests that higher AI literacy levels correlate with increased dependence on AI tools, raising concerns about excessive reliance on AI-driven educational technologies (Wijaya et al., 2024). These findings underscore the need for a balanced approach to AI literacy that integrates technical, ethical, and pedagogical frameworks for responsible AI use in education without compromising essential cognitive and social skills.

Demographic variables, such as gender, teaching discipline, and educational background, have also been examined in relation to AI literacy. Research findings indicate no significant differences in AI literacy between male and female teachers, both before and after participating in professional development, suggesting that AI literacy training is equally effective for both genders and reinforcing the importance of inclusive AI education programs (Younis, 2024).

However, subject specialization has been found to significantly impact the AI literacy levels, with teachers from computer science backgrounds exhibiting the highest AI literacy scores, while those in fields such as religious studies demonstrated lower pre-intervention AI literacy levels (Ding et al., 2024b). Notably, structured AI literacy training effectively mitigated these disciplinary disparities, leading to comparable AI literacy levels across different specializations post-intervention (Ding et al., 2024b).

Additionally, educational background plays a crucial role in the development of AI literacy, as teachers with higher academic qualifications tend to have a more advanced understanding of AI concepts, applications, and ethical considerations (Zhao et al., 2022).

Professional development has proven effective in bridging knowledge gaps across varying educational levels, underscoring the need for AI literacy training to be integrated into both initial teacher education and ongoing professional development programs (Hur, 2025).

Regarding age, there was no significant direct correlation between teachers' age and their level of AI literacy. However, younger teachers, particularly those under 40, tended to demonstrate higher confidence in AI adoption, likely due to their greater familiarity with digital technologies. Older teachers, while not necessarily having lower AI literacy scores, expressed more skepticism regarding AI integration in education, citing concerns about pedagogical disruptions and ethical implications (Wijaya, et al., 2024).

Furthermore, AI literacy levels tend to vary according to career stages, with novice and mid-career educators demonstrating moderate AI literacy, whereas experienced educators often exhibit more advanced AI knowledge and application skills (Su, 2024; Zhao et al., 2022).

Another study found that teachers who actively engaged with AI tools demonstrated higher AI literacy, as frequent usage facilitated greater familiarity with AI concepts, applications, and ethical considerations (Wijaya et al., 2024). These findings underscore the importance of ongoing AI training to enable teachers at all career stages to effectively integrate AI into their pedagogical practices.

This study explores the associations between teachers' AI literacy and several variables, drawing on recent research identifying links between AI literacy and various variables (Deshen & Aharony, 2024, 2025; Due et al., 2024). By examining the intersection of

AI literacy with (1) AI acceptance, (2) computational thinking, (3) AI anxiety, and (4) the Digital divide, we seek to gain a comprehensive understanding of the variables associated with teachers' AI literacy. The results will guide the creation of focused professional development programs, inform the enhancement of teacher preparation curricula, and support evidence-based policymaking to improve AI literacy in education.

AI acceptance

The adoption of AI technology has evolved from earlier frameworks, such as the Technology Acceptance Model (TAM) formulated by Davis (1985) and subsequently refined by Venkatesh and Davis (2000), as well as the Unified Theory of Acceptance and Use of Technology (UTAUT) proposed by Venkatesh et al. (2012). These models predominantly examine behavioral intentions and the actual utilization of novel technologies. Furthermore, the Artificially Intelligent Device Use Acceptance (AIDUA) model, introduced by Gursoy et al. (2019), significantly contributes to this discourse by concentrating specifically on AI technologies. This study utilizes six variables from the AIDUA scale: social influence, hedonic motivation, performance expectancy, effort expectancy, willingness to use AI, and positive emotions about AI. These variables were selected based on previous studies (e.g., Ma & Huo, 2023; Strzelecki, 2023; Strzelecki & ElArabawy, 2024) that demonstrate associations between AI acceptance and AI usage. Table 1 presents the AIDUA model variables, their definitions, and their relationships with TAM, UTAUT, and UTAUT2.

Research on *social influence* and its impact on the acceptance and use of AI technologies has produced mixed findings. While some studies report a positive correlation (Gursoy et al., 2019; Strzelecki, 2023; Strzelecki & ElArabawy, 2024; Wang et al., 2023), others find no significant relationship (Bhat et al., 2024; Cabrera-Sánchez et al., 2021). However, social influence plays a key role in shaping students' intentions to adopt

Table 1 The Artificially Intelligent Device Use Acceptance (AIDUA) variables

	Definition	TAM (Davis, 1989)*	UTAUT (Venkatesh et al., 2003)
Social influence	The perception that family, peers, or community expect and encourage the individual to use AI	✗	✓
Hedonic motivation	The sense of enjoyment and pleasure experienced when interacting with AI tools	✗	✗
Performance expectancy	The belief that AI will improve effectiveness, efficiency, or daily outcomes	✓[1]	✓
Effort expectancy	The degree to which using AI is seen as easy and requires little effort	✓[2]	✓
Willingness to use AI	The individual's readiness to adopt AI technologies in different contexts	✓[3]	✓[3]
Positive emotion toward AI	Favorable affective responses such as trust, comfort, or excitement that promote engagement with AI	✗/✓[4]	✗

[1] Originally named perceived usefulness

[2] Originally named perceived ease of use

[3] Originally named perceived behavioral intention

[4] Emotion is only partly reflected in TAM (as attitude toward use) and in UTAUT2 extensions (as enjoyment / trust) but is not included as a core construct

generative AI for learning, with a stronger effect observed among female students than males (Elshaer et al., 2024; Gc et al., 2024). Additionally, students of social sciences demonstrate a higher susceptibility to social influence in AI adoption than those in the natural sciences, underscoring disciplinary differences (Elshaer et al., 2024).

Hedonic motivation is strongly associated with AI usage (Cabrera-Sánchez et al., 2021; Chi et al., 2022; Strzelecki, 2023; Wang et al., 2023) and correlates with AI literacy levels (Deshen & Aharony, 2024, 2025). It is a key factor associated with the intention to use GenAI and enhance user experience among educators (Bhat et al., 2024). Although generally significant with AI adoption, a particular study indicated that it had no evident effect on students' intentions to utilize ChatGPT in educational settings (Gc et al., 2024).

Performance expectancy has been associated with the intention to use Web 2.0 technologies (Aharony & Prebor, 2015; Aharony, 2014, 2015a, 2015b), ChatGPT (Strzelecki, 2023), and AI service robots (Lu et al., 2019). It was significantly associated with AI adoption in higher education (Strzelecki & ElArabawy, 2024) and students' behavioral intention to use AI for academics, with a stronger impact on male students than females (Elshaer et al., 2024). Among educators, performance expectancy promotes its integration into teaching (Bhat et al., 2024). However, some studies found no significant correlation between performance expectancy and AI adoption among university students (Gc et al., 2024) or teachers (Al-Abdullatif, 2024).

Effort expectancy is also a key factor in technology adoption, associated with the use of Web 2.0 technologies (Aharony & Prebor, 2015; Aharony, 2014, 2015a, 2015b), mobile learning (Hu et al., 2020), and learning management systems (Jakkaew & Hemrungrote, 2017). It is significantly associated with AI adoption in higher education (Strzelecki & ElArabawy, 2024) and is a strong predictor of behavioral intention and usage of AI across various industries (Kelly et al., 2023). Research also links effort expectancy to students' intention to use GenAI (Elshaer et al., 2024). Among educators, it plays a crucial role in adopting ChatGPT, shaping their willingness to use it (Al-Abdullatif, 2024; Bhat et al., 2024).

The *willingness to use AI* is also linked to AI literacy levels (Deshen & Aharony, 2024, 2025; Due et al., 2024), and AI usage in higher education (Strzelecki & ElArabawy, 2024). Willingness to use AI has been identified as a key factor in AI adoption (Vitezić & Perić, 2021), while age also plays a role, with younger users being more inclined to embrace AI technologies like ChatGPT (Ma & Huo, 2023). Among educators, the willingness to use AI is a strong predictor of actual ChatGPT adoption (Bhat et al., 2024).

Positive emotions increase the possibility of acknowledging AI, whereas negative emotions can foster resistance to it (Gursoy et al., 2019; Lu et al., 2019). Furthermore, a positive emotional state is linked to customers' willingness to use AI (Gursoy et al., 2019). Additionally, positive emotions and levels of AI literacy are significantly associated (Deshen & Aharony, 2024, 2025).

Therefore, we hypothesized the following:

H1a. The higher the teachers' scores of AI Acceptance (social influence, hedonic motivation, performance expectancy, willingness to use AI, and positive emotions), the higher their AI literacy.

H1b. The lower the teachers' scores of effort expectancy, the higher their AI literacy.

Computational thinking

Computational thinking, introduced by Seymour Papert, emphasizes the use of computers as powerful tools for learning and problem-solving across various domains. This perspective established computational thinking as a fundamental cognitive skill with broad applications (Papert, 1993). It is a problem-solving approach that uses digital tools encompassing algorithmic thinking, data analysis, and pattern recognition (ISTE, 2015). Wing (2006, 2011) defined computational thinking as a universally applicable cognitive skill for understanding and solving complex problems. Recognizing its importance, researchers have focused on integrating computational thinking into educational curricula, aiming to prepare students for 21st-century challenges (Yadav et al., 2016). This comprehensive approach enhances problem-solving skills and prepares individuals to effectively engage in the increasingly digital and complex world.

Regarding AI usage, Yilmaz and Yilmaz (2023) found that using GenAI, particularly ChatGPT, significantly improves computational thinking skills among programming students. Regarding AI literacy, Celik (2023) suggests that computational thinking is essential for developing AI literacy, enhancing understanding of AI functions and technologies, and sharpening critical evaluation skills for assessing AI outcomes and their ethical implications. Other researchers (e.g., Abar et al., 2024) have emphasized that computational thinking and AI literacy rely on problem-solving and critical thinking skills, meaning that developing computer thinking skills simultaneously builds AI literacy, creating a reciprocal relationship that prepares students for an increasingly technology-driven world. Both skills have become increasingly essential as technology advances, highlighting the need for further research into their relationship.

Therefore, we hypothesized the following:

H2. The higher the teachers' scores in computational thinking (creativity, algorithmic thinking, cooperation, critical thinking, and problem-solving), the higher their AI literacy.

AI anxiety

Technology anxiety refers to an individual's psychological apprehension and unease regarding their capability and willingness to use technology-related tools. It encompasses feelings of apprehension, fear, and discomfort associated with using technology, which can affect an individual's behavior and attitudes toward technological tools (Meuter et al., 2003). Computer anxiety is defined as the fear that individuals experience while using computers. (Heinssen et al., 1987). This encompasses negative feelings about actual and anticipated computer interactions, often stemming from cognitive and psychological variables. This anxiety can lead to avoidance behaviors, whereby people may completely shy away from using computers (Heinssen et al., 1987; Khasawneh, 2018). As technology has evolved rapidly, anxieties primarily associated with general technology and computer use have expanded to encompass more advanced systems, particularly AI.

AI anxiety refers to the feelings of fear and uneasiness that individuals may encounter regarding AI technologies (Li & Huang, 2020) and is distinguished from other forms of anxiety by its unique components (Wang & Wang, 2022). AI anxiety reflects broader societal concerns about the rapid advancement and potential consequences of AI technologies, including ethical dilemmas, privacy issues, and socioeconomic inequalities.

Understanding AI anxiety is crucial, as it is significantly associated with individuals' attitudes toward AI and their willingness to engage with it (Kaya et al., 2024). Key concerns include privacy violations, algorithmic bias, job displacement, and the challenge of keeping up with rapid technological changes (Li & Huang, 2020). Other concerns include existential risks to humanity, ethical dilemmas in AI decision making, and the potential for artificial consciousness (Li & Huang, 2020). The lack of transparency in AI systems heightens these anxieties as people grapple with the implications of autonomous technologies for society and human life. These concerns reflect broader societal apprehensions about the impact of AI on privacy, employment, ethics, and human agency in an increasingly AI-driven world (Li & Huang, 2020). Studies exploring AI anxiety have shown that women exhibit higher anxiety levels than men (Hopcan et al., 2024; Kaya et al., 2024). Furthermore, job replacement concerns, implementation uncertainty, concerns about the ethical implications of AI, limited exposure and understanding, and limited AI usage experience are associated with AI anxiety (Hopcan et al., 2024). Studies found that higher AI literacy was associated with lower levels of AI anxiety among the general population (Schiavo, et al., 2024) and among university students (Cengiz & Peker, 2025). Despite the growing integration of AI in education, the specific correlation between teachers' anxiety about AI and their AI literacy remains underexplored.

Based on the above-mentioned studies, we hypothesized the following:

H3. The higher the teachers' scores in AI anxiety, the lower their AI literacy.

Digital divide

The digital divide refers to the gap in access to and use of digital technologies and the Internet. This reflects broader social inequalities and requires strategies to improve access and digital skills for all (Scheerder et al., 2017). The digital access gap is not just about who has access and who does not. It is also concerned with differences in expertise, usage, and the time taken for people to adopt new technologies. Research has shown that variables such as age, gender, ethnicity, and socioeconomic status play a significant role in these disparities (Van Dijk, 2006).

A systematic literature review (Scheerder et al., 2017) revealed three levels of the digital divide, shifting from a first-level digital divide (referring to physical access) to a second-level digital divide (referring to skills and usage), to a third-level digital divide (referring to tangible outcomes). The first level highlights the gap between those with and without Internet connectivity, which is often associated with variables such as geographic location, socioeconomic status, and infrastructure availability (Jung et al., 2001; Segev & Ahituv, 2010; Van Dijk, 2020).

A previous study (Celik, 2023) suggested that the digital divide is closely linked to AI literacy, as individuals with better access and those skilled in using information and communication technologies, as well as AI-based technologies, tend to have higher levels of AI literacy. Celik (2023) emphasizes the significant role of digital inequalities in shaping AI literacy and underscores the importance of interventions to address different aspects of the digital divide to promote AI literacy across diverse populations.

Bridging the digital divide is essential for enabling equitable AI usage, as enhancing reliable and affordable internet connectivity in low-income and lower-middle-income countries is crucial for empowering users to fully engage with advanced tools like

ChatGPT (Khowaja et al., 2024). Specifically, regarding teachers, the correlation between teachers' Digital divide and their AI literacy remains underexplored.

Based on the above-mentioned studies, we hypothesized the following:

H4. The higher the teachers' scores in the digital divide, the higher their AI literacy.

Based on the research hypotheses, our research model (Fig. 1) examines the relationships among these variables to provide a comprehensive understanding of how they interact.

Research gap and study objectives

Despite growing scholarly attention to artificial intelligence (AI) in education, research on teachers' AI literacy remains fragmented and underdeveloped. Much of the existing literature conceptualizes AI literacy broadly, focusing on general populations, students, or non-educational contexts (Cox, 2024; Long & Magerko, 2020; Ng et al., 2021). Studies addressing teachers have primarily explored their attitudes toward AI (Al-Abdullatif, 2024; Bhat et al., 2024) or the effectiveness of short-term training interventions (Ding et al., 2024b; Younis, 2024), leaving a critical gap in understanding the multifaceted nature of teachers' AI literacy and its relationship with other psychological, cognitive, and contextual variables. Furthermore, while frameworks such as UNESCO's (2024) global call for AI literacy underscore its ethical and social importance, empirical work has not yet fully examined the determinants of teachers' AI literacy in everyday professional practice.

Existing studies also tend to treat AI literacy as a discrete skill set rather than as a construct embedded in broader technological, cognitive, and affective dimensions. For instance, prior research has established the relevance of AI acceptance factors such as hedonic motivation, effort expectancy, and willingness to use AI for predicting adoption (Gursoy et al., 2019; Strzelecki & ElArabawy, 2024), but few have examined how these variables translate into measurable literacy outcomes among teachers. Similarly, computational thinking has been identified as a conceptual and pedagogical foundation for AI literacy (Celik, 2023; Yadav et al., 2016), yet empirical evidence linking these constructs in the teaching profession remains scarce. Additionally, the potential negative influences of AI anxiety (Kaya et al., 2024; Wang & Wang, 2022) and the persistent impact of the digital divide on equitable access to AI tools (Celik, 2023; Scheerder et al., 2017) have

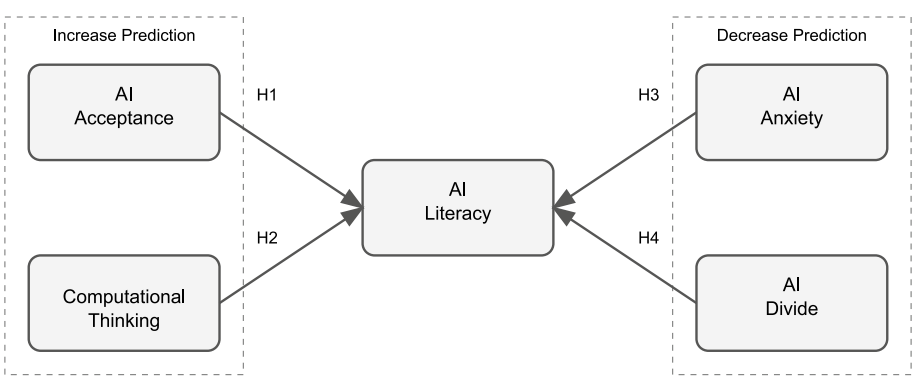

Fig. 1 Hypothesis research model

received limited investigation in teacher populations, especially in contexts outside East Asia and North America.

The present study seeks to address these gaps by offering a comprehensive examination of the variables associated with teachers' AI literacy within the Israeli educational system. Specifically, it investigates how factors of AI acceptance, computational thinking, AI anxiety, and the Digital divide interact to explain teachers' AI literacy levels. By integrating theoretical insights from technology acceptance (Venkatesh et al., 2012), computational thinking (Wing, 2011), and digital inequality research (Van Dijk, 2020), this study aims to contribute an empirically grounded model of AI literacy that reflects both the cognitive and affective dimensions of teachers' engagement with AI technologies. The study's objectives are therefore twofold: (1) to identify the key predictors that significantly contribute to teachers' AI literacy and (2) to provide evidence-based insights to inform professional development, curriculum design, and policy initiatives promoting equitable and responsible AI integration in education.

Methods

Research process

This study employed a quantitative research method. The questionnaire consisted of six sections: a background survey and five validated scales that addressed AI literacy, AI acceptance, computational thinking, AI anxiety, and the Digital divide.

The study was conducted among Israeli teachers. All survey respondents were recruited through an online consumer panel provided by "Panel4all" (https://www.panel4all.co.il) in Hebrew. The questionnaire was designed to be anonymous. Before completing the survey, informed consent was obtained from every participant. In line with the platform's policy and agreement with participants, they received compensation for their participation. The survey was anonymous, and no personal identification information was collected. The institute's ethics committee approved the study before its commencement.

Participants

The sample consisted of 270 teachers from Israel, aged 25 to 78 years (M = 42.08, SD = 12.86). Approximately two-thirds of the participants were female (n = 188, 69.6%). Most participants held a bachelor's degree (n = 179, 66.3%), while the remaining participants had a master's degree or higher (n = 91, 33.7%). About one-sixth of the teachers taught Science and Technology (n = 45, 16.7%), while the majority taught subjects such as math, language, and arts (n = 225, 83.3%). Regarding the use of AI, most respondents reported using it (n = 196, 72.6%), with approximately half indicating that they used AI for work purposes (n = 138, 51.1%).

Measures

This study uses five validated measurement tools to evaluate the key variables, along with background data and is fully represented in Appendix 1. The background data collected from participants included personal background variables, and characteristics of GenAI chatbot usage, such as whether they use AI for work.

Each of the five validated measurement instruments consists of multiple subscales that have demonstrated reliability:

1. Several validated instruments have been developed in recent years to assess artificial intelligence (AI) literacy, including teacher-specific scales (Chiu, 2024; Ning et al., 2025) and general measures adapted for educational contexts (Ding et al., 2024a, 2024b; Wang et al., 2022). Among these, the present study employed the AI Literacy Scale developed by Wang et al. (2022), which operationalizes the four key dimensions identified by Ng et al. (2021): knowing and understanding AI, using and applying AI, evaluating and creating AI, and AI ethics. This instrument was selected because it provides a psychometrically validated and reliable measure (Cronbach's $\alpha = .93–.95$ in the original study) suitable for non-technical adult users, and it aligns closely with leading conceptual frameworks (Chiu et al., 2024; Long & Magerko, 2020; Ng et al., 2021). The scale enables empirical evaluation of the four widely accepted dimensions of AI literacy while maintaining a concise and clear structure, making it well-suited for adaptation to the Israeli teacher population examined in this study. The items were measured using a 5-point Likert scale (1 = no AI literacy to 5 = complete AI literacy). The current study demonstrated excellent internal consistency ($\alpha = .95$), confirming its suitability for the teacher population examined.
2. Six subscales of the AI acceptance AIDUA model (Gursoy et al., 2021): (1) social Influence, (2) hedonic motivation, (3) performance expectancy, (4) effort expectancy, (5) willingness to use AI, and (6) emotions towards AI. These variables were selected based on their established correlations in previous studies (Deshen & Aharony, 2024, 2025). All items were measured using a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). One exception was the section used to assess emotions, as it was measured using a 5-point bipolar scale (e.g., 1 = bored; 5 = relaxed). Cronbach's alpha values for the six subscales in the current study were 0.92, 0.93, 0.89, 0.76, 0.88, and 0.92, respectively.
3. The Computational thinking model (Korkmaz et al., 2019) includes five subscales: (a) creativity, (b) algorithmic thinking, (c) cooperativity, (d) critical thinking, and (e) problem-solving. All items were measured using a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). Cronbach's alpha values for the five subscales in the current study were 0.84, 0.91, 0.88, 0.85, and 0.81, respectively.
4. The AI anxiety model (Wang & Wang, 2022) refers in one scale to learning, job replacement, sociotechnical blindness, and AI configuration. The Cronbach's alpha for the entire questionnaire in the current study was 0.95.
5. The physical access sub-scale of the Digital divide model (Soomro et al., 2018), with Cronbach's alpha in the current study of 0.80.

Table 2 outlines the measures used in this study and their subscales.

Statistical analysis

Before conducting the subsequent analyses, the study variables were scrutinized for outliers and normality distribution. Following this, associations between background variables and study variables were examined. Pearson's correlations were used for continuous

Table 2 Measures of the study

Dependent variable	Independent variables			
AI Literacy	**AI acceptance**	**Computational thinking**	**AI anxiety**	**Digital divide**
(AILS)	(AIDUA)			
	Social influence Hedonic motivation Performance expectancy Effort expectancy Willingness to use Positive emotions	Creativity Algorithmic thinking Cooperativity Critical thinking Problem-solving		Physical access
Wang et al. (2023)	Gursoy et al. (2019)	Korkmaz et al. (2017)	Wang and Wang (2022)	Soomro et al. (2018)

variables. Additionally, independent sample t-tests were performed to compare IA literacy across dichotomous variables such as gender, education, teaching area, and the use of AI for work. Then, hierarchical multiple regression analysis was conducted to predict AI literacy scores. The first five steps utilized the Enter method for variable entry, while the last step utilized the Stepwise method for entering interactions. The Stepwise method was used only in the final step to identify significant interaction terms while maintaining model parsimony. Although stepwise procedures may increase the risk of overfitting (Babyak, 2004), this exploratory approach was considered appropriate for identifying potential moderating effects. The findings are therefore interpreted with due caution. In addition to R^2 and ΔR^2, Cohen's f^2 was calculated to estimate the effect size of each hierarchical block, based on the change in R^2 between consecutive models (Cohen, 1988). Values of .02, .15, and .35 were interpreted as small, medium, and large effects, respectively. Prior to analysis, all continuous variables were mean centered to enhance the interpretability of results (Darlington & Hayes, 2017; Hayes, 2022). Additionally, Variance inflation factor (VIF) values were inspected to detect multicollinearity issues (Hair et al., 2014). The analyses were conducted using IBM SPSS statistics version 29.

Findings

No outliers were identified (Z scores $<|4|$) (Hair et al., 2014), and all variables exhibited values within the normal range (Skewness$|<1$ and $|$Kurtosis$|<4$) (Kline, 2016).

Descriptive statistics and associations

Table 3 presents the descriptive statistics and associations for background variables related to AI literacy.

Regarding the background variables, education level was associated with AI literacy, with teachers with a master's degree or higher scoring higher than those with a bachelor's degree. Additionally, AI literacy was significantly higher among Science and Technology teachers than among those teaching math, language, or arts. Furthermore, the use of AI for work was associated with higher AI literacy scores compared to those who do not use AI for work. Conversely, no significant differences in AI literacy were found in terms of gender or age.

Table 3 Means, standard deviations, and associations for background variables

		AI literacy				
		M	**SD**	**t**	**df**	**r**
Age		42.09	12.86			−0.10
Gender	Male	3.49	0.61	0.00	268.00	
	Female	3.49	0.67			
Education	Bachelor	3.42	0.63	**−2.24***	268.00	
	Master's degree or higher	3.61	0.67			
Teaching area	Science and Technology	3.72	0.72	**−2.72****	268.00	
	Other	3.43	0.63			
Use of AI for work	Yes	3.74	0.56	**−7.21*****	268.00	
	No	3.22	0.64			

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4 presents the descriptive statistics and associations of the study variables related to AI literacy.

The AI Acceptance sub-scales social influence, hedonic motivation, performance expectancy, willingness to use AI, and emotions toward AI were positively associated with AI literacy. From the Computational Thinking scale, creativity, algorithmic thinking, cooperativity, and critical thinking were also positively associated with AI literacy. However, problem-solving did not. Additionally, digital divide was found to positively correlate with AI literacy, indicating that teachers with better access to information and communication technologies tend to exhibit higher levels of AI literacy. Conversely, as expected, effort expectancy and AI anxiety were negatively associated with AI literacy.

The analysis of correlations among the six sub-scales of AI acceptance model reveals a complex network, with hedonic motivation notably demonstrating the strongest correlation with the other variables. Social influence has moderate positive correlations, especially with performance expectancy and hedonic motivation. Effort expectancy shows predominantly weak or non-significant correlations, only significantly correlating with social influence.

Analyzing the interrelationships among the five sub-scales of the computational thinking model shows that critical thinking has the strongest correlations with creativity, algorithmic thinking, and cooperativity. Cooperativity has moderate correlations with creativity and algorithmic thinking. Problem-solving shows weak positive correlations with creativity and critical thinking, no significant correlation with algorithmic thinking, and a weak positive correlation with cooperativity. Algorithmic thinking and creativity have a moderate correlation.

More findings point out that AI anxiety shows a moderate positive correlation with effort expectancy, digital divide shows moderate positive correlations with hedonic motivation, and problem-solving shows an unexpected pattern of negative correlations with several acceptance variables, including social influence, performance expectancy, and effort expectancy.

Table 4 Means, standard deviations, and associations for study variables

			M	SD	AI literacy	2	3	4	5	6	7	8	9	10	11	12	13
	1	AI literacy	3.49	0.65	—												
AI acceptance	2	Social Influence	2.65	1.09	**0.23*** **	—											
	3	Hedonic Motivation	3.50	1.04	**0.59*** **	**0.47*** **	—										
	4	Performance Expectancy	3.02	1.00	**0.30*** **	**0.50*** **	**0.52*** **	—									
	5	Effort Expectancy	2.54	0.93	**−0.17** **	**0.35*** **	0.03	**0.29*** **	—								
	6	Willingness to accept the use of AI	3.52	1.05	**0.58*** **	**0.28*** **	**0.66*** **	**0.37*** **	−0.06	—							
	7	Positive Emotions	3.38	0.98	**0.42*** **	**0.31*** **	**0.49*** **	**0.34*** **	0.03	**0.50*** **	—						
Computational thinking	8	Creativity	3.92	0.61	**0.27*** **	0.04	**0.26*** **	0.11	−0.11	**0.35*** **	**0.34*** **	—					
	9	Algorithmic thinking	3.08	1.00	**0.15* **	**0.40*** **	**0.32*** **	**0.26*** **	**0.21*** **	0.09	**0.26*** **	**0.24*** **	—				
	10	Cooperativity	3.69	0.92	**0.16** **	**0.16** **	**0.25*** **	0.02	−0.02	**0.28*** **	**0.26*** **	**0.31*** **	**0.24*** **	—			
	11	Critical thinking	3.70	0.76	**0.28*** **	**0.18** **	**0.33*** **	**0.16** **	−0.03	**0.32*** **	**0.3*** **	**0.56*** **	**0.47*** **	**0.45*** **	—		
	12	Problem − solving	3.34	0.81	0.11	**−0.28*** **	−0.02	**−0.22*** **	**−0.39*** **	0.09	0.02	**0.18** **	−0.04	**0.12* **	**0.16** **	—	
	13	AI anxiety	3.74	1.26	**−0.25*** **	0.08	−0.11	0.04	**0.31*** **	**−0.13* **	**−0.21*** **	−0.05	**−0.15* **	0.02	−0.11	**−0.27*** **	—
	14	Digital divide	12.37	4.60	**0.27*** **	0.10	**0.21*** **	0.09	−0.04	**0.23*** **	**0.19** **	**0.15* **	0.03	**0.14* **	**0.13* **	**0.12* **	−0.06

$N = 270$. Pearson's coefficients are presented. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

Variables contributing to the explanation of the variance of teacher's AI literacy

Table 5 presents the results of the hierarchical multiple regression analysis. In Step 1, background variables (age, gender, education, teaching area, and AI usage for work) were entered. In Step 2, the six sub-scales of the AIDUA model (social influence, hedonic motivation, willingness to use AI and emotions toward AI) were included. Step 3 incorporated the five sub-scales of the computational thinking model (creativity, algorithmic thinking, cooperativity, critical thinking, and problem-solving). Steps 4 and 5 included AI anxiety and digital divide, respectively. In the final step, interactions were entered. The ENTER method was utilized for variable entry in Steps 1 to 5, while the Stepwise method was employed for entering interactions in the final step.

The first step introduced the background variables, accounting for 20% of the variance in AI literacy ($f^2 = 0.25$, medium-to-large effect). Age, gender, and education level were not significant predictors. However, the teaching subject emerged as a significant predictor, with Science and Technology teachers exhibiting slightly higher AI literacy than those teaching math, language, or arts. The strongest predictor in this step was AI usage for work, indicating that individuals who use AI for work have significantly higher AI literacy scores.

In the second step, the six variables from the AI acceptance construct were introduced, adding an extra 28% to the explained variance of AI literacy ($f^2 = 0.54$, large effect); in particular, hedonic motivation and the willingness to use AI were the contributors to the variance explanation. These results highlight that AI acceptance is a strong and significant predictor of AI literacy.

In the third step, the five variables from the computational thinking construct were introduced ($f^2 = 0.02$, small effect), followed by AI anxiety in the fourth step ($f^2 = 0.00$, negligible) and digital divide in the fifth step ($f^2 = 0.02$, small effect). Interestingly, none of these newly added variables significantly contributed to the explained variance of AI literacy.

The sixth step examined the contributions of the interactions between education × effort expectancy and gender × teaching area. A significantly small contribution was identified. These interactions added another 2% to explaining the variance in AI literacy ($f^2 = 0.04$, small effect).

The interaction between education and effort expectancy was found to be significant and negative. Figure 2 illustrates that among teachers with a master's degree or higher, there was a much stronger relationship between effort expectancy and AI literacy compared to teachers with a bachelor's degree. This indicates that, for teachers with master's degrees or higher, higher effort expectancy scores are linked to lower AI literacy than they are for teachers with bachelor's degrees.

The interaction between gender and teaching area was found to be significant and positive. Figure 3 illustrates that for men, the teaching area has a significant impact on AI literacy. Specifically, men who teach Science and Technology have a higher expected AI literacy score compared to men who do not teach Science and Technology. However, for women, teaching science and technology did not show a significant association with AI literacy when compared to those teaching other subjects. These findings suggest that the teaching area (Science and Technology vs. Arts, Language,

Table 5 Results of hierarchical multiple regression analysis predicting AI literacy

Variable	Model 1			Model 2			Model 3			Model 4			Model 5			Model 6		
	B	*SE B*	B	*B*	*SE B*	β	*B*	*SE B*	β	*B*	*SE B*	β	*B*	*SE B*	β	*B*	*SE B*	β
Intercept	3.38	0.06		3.18	0.06		3.36	0.05		3.35	0.05		3.35	0.05		3.37	0.05	
Gender	−0.06	0.08	−0.04	−0.04	0.06	−0.03	−0.03	0.07	−0.02	−0.04	0.07	−0.03	−0.04	0.07	−0.03	−0.09	0.07	−0.06
Age	−0.01	0.00	−0.11	0.00	0.00	−0.07	0.00	0.00	−0.08	0.00	0.00	−0.06	0.00	0.00	−0.07	0.00	0.00	−0.09
Education	0.10	0.08	0.07	0.06	0.07	0.04	0.05	0.07	0.04	0.04	0.07	0.03	0.02	0.07	0.01	0.01	0.07	0.00
Teaching area	0.23	0.10	0.13*	0.09	0.08	0.05	0.08	0.08	0.05	0.08	0.08	0.05	0.09	0.08	0.05	−0.03	0.10	−0.02
Use of AI for work	0.49	0.07	0.38***	0.20	0.07	0.16**	0.24	0.07	0.18***	0.23	0.07	0.18***	0.20	0.07	0.16**	0.20	0.07	0.15**
Social Influence				0.00	0.03	−0.01	0.00	0.04	0.00	0.01	0.04	0.01	0.00	0.04	0.01	0.01	0.04	0.02
Hedonic Motivation				0.20	0.04	0.31***	0.19	0.05	0.30***	0.19	0.05	0.30***	0.18	0.05	0.29***	0.18	0.04	0.29***
Performance Expectancy				0.02	0.04	0.04	0.02	0.04	0.03	0.02	0.04	0.03	0.02	0.04	0.04	0.01	0.04	0.02
Effort Expectancy (EE)				−0.10	0.04	−0.15**	−0.09	0.04	−0.14*	−0.08	0.04	−0.11*	−0.08	0.04	−0.11*	−0.04	0.04	−0.05
Willingness I				0.15	0.04	0.24***	0.14	0.04	0.23***	0.14	0.04	0.22**	0.14	0.04	0.22**	0.13	0.04	0.22**
Positive Emotions				0.06	0.04	0.08	0.04	0.04	0.06	0.03	0.04	0.04	0.03	0.04	0.04	0.03	0.04	0.05
Creativity							0.07	0.06	0.06	0.08	0.06	0.07	0.07	0.06	0.06	0.06	0.06	0.06
Algorithmic thinking							0.00	0.04	0.00	−0.01	0.04	−0.02	−0.01	0.04	−0.01	−0.02	0.04	−0.04
Cooperativity							−0.04	0.04	−0.05	−0.03	0.04	−0.04	−0.03	0.04	−0.04	−0.02	0.04	−0.03
Critical thinking							0.05	0.06	0.06	0.05	0.06	0.06	0.05	0.05	0.06	0.07	0.06	0.08
Problem − solving							0.01	0.04	0.01	−0.01	0.04	−0.01	−0.01	0.04	−0.01	−0.02	0.04	−0.02
AI anxiety										−0.05	0.03	−0.10	−0.05	0.03	−0.10	−0.05	0.03	−0.09
Digital divide													0.01	0.01	0.09	0.01	0.01	0.08
Education x EE																−0.15	0.07	−0.13*
Gender x Teaching area																0.37	0.17	0.14*
R^2		.20***			.48***			.49***			.49***			.50***			.52***	
ΔR^2		.14***			.28***			.01			.00			.01			.02*	
Cohen's f^2 (block)		.25			.54			.02			.00			.02			.04	

$N = 270$. All independent continuous variables are mean-centered. *$p < .05$. **$p < .01$. ***$p < .001$. Cohen's f^2 indicates each block's effect size, computed from the change in R^2 between consecutive models (Cohen, 1988). .02 = small, .15 = medium, .35 = large

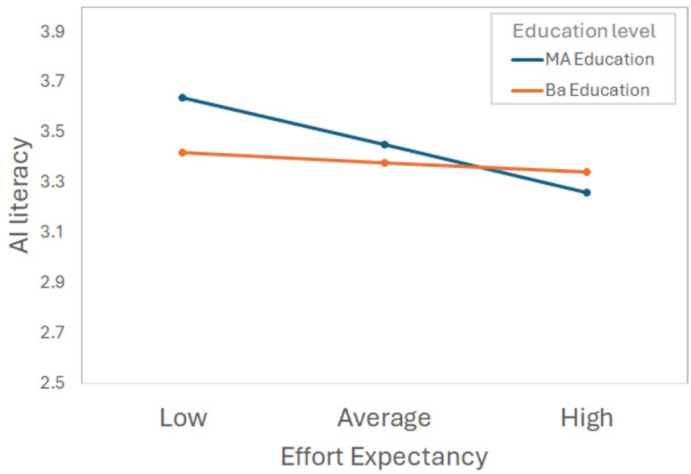

Fig. 2 The effect of Education and Effort Expectancy on AI Literacy. *Note.* N = 270. The predicted AI literacy scores for individuals with bachelor's degrees and master's degrees or higher education levels across varying levels of effort expectancy (low, average, high). Mean-centered values are presented for Effort Expectancy

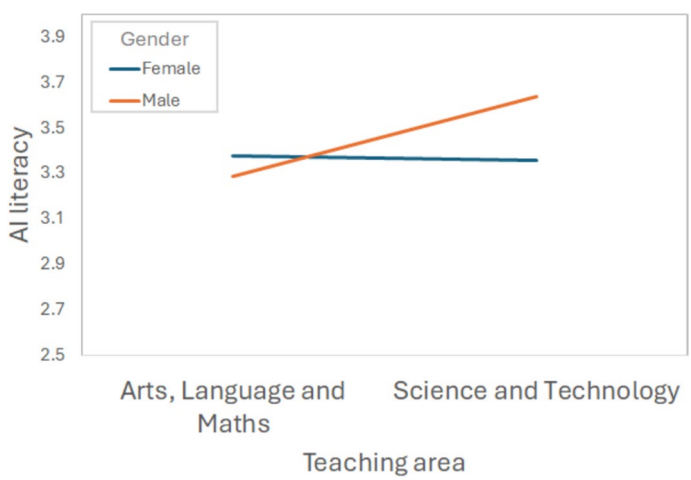

Fig. 3 The effect of Gender and Teaching Area on AI Literacy. *Note* N = 270. The predicted AI literacy scores for male and female teachers across teaching areas (Arts, Language, and Math vs. Science and Technology)

and Math) is significantly associated with AI literacy among men, while it is not a significant predictor of AI literacy for women.

Finally, there were no multicollinearity issues (all VIF values < 2.00), and the model explained about 52% of the variance in AI literacy scores.

Discussion

The present study contributes to the growing body of research on teachers' Artificial Intelligence (AI) literacy by examining its associations with various variables. The findings demonstrate that attitudinal and motivational variables, particularly hedonic motivation and willingness to use AI, emerge as the strongest predictors of AI literacy, surpassing technical or demographic variables.

H1a was accepted, affirming that the five predicted sub-scales of AI acceptance were associated with AI literacy. Specifically, Hedonic motivation emerged as the strongest predictor of AI literacy among all components of AI acceptance. The finding that *hedonic motivation* is the strongest predictor of AI literacy aligns with studies demonstrating the centrality of enjoyment and perceived pleasure in AI use among educators and students (Cabrera-Sánchez et al., 2021; Chi et al., 2022; Strzelecki, 2023). Similarly, other reserchers (Deshen & Aharony, 2025) reported that hedonic motivation and willingness to use AI were the strongest variables explaining librarians' AI literacy. However, Gc et al., (2024) showed no clear effect on students' intentions to use ChatGPT in educational settings. These differing results might reflect cultural and institutional differences across countries and suggest the need for more evidence in future research.

Additionally, the willingness to use AI was also among the strongest of the AI acceptance components, meaning that teachers who expressed greater readiness to adopt and implement AI technologies demonstrated higher levels of AI literacy, making the willingness to use AI one of the key predictors alongside hedonic motivation. This finding extends prior research showing that behavioral intention is central to AI acceptance (Bhat et al., 2024; Vitezić & Perić, 2021). Teachers who demonstrate willingness toward AI usage, likely engage in self-initiated exploration that increases both conceptual understanding and ethical awareness, leading to higher literacy.

H1b was accepted, affirming that effort expectancy was negatively correlated with AI literacy, meaning that teachers who perceived AI as requiring less effort to use demonstrated higher levels of AI literacy. Previous studies revealed similar significant negative correlations between effort expectancy and technology adoption in higher education (Strzelecki & ElArabawy, 2024), and other industries (Kelly et al., 2023). Additionally, our findings support Al-Abdullatif's (2024) and Bhat et al.'s (2024) conclusions about effort expectancy's crucial role in educators' willingness to use AI technologies like ChatGPT.

The analysis revealed that the link between effort expectancy and AI literacy varies by educational background. Teachers with master's degrees or higher exhibit a stronger negative relationship than those with bachelor's degrees, suggesting that these teachers may find AI more complex, and this perception is hindering their development of AI literacy. Interestingly, the interaction between effort expectancy and education showed an unexpected pattern that goes against previous research, which typically links higher educational levels with greater confidence in technology and less perceived effort in adopting new technology (Venkatesh et al., 2012). One possible explanation is that more highly educated teachers tend to have stronger analytical and metacognitive skills, which increase their awareness of AI's conceptual and ethical complexities. Another explanation might be that their deeper cognitive engagement with the technology raises perceived mental load and makes AI use seem more demanding. This pattern suggests that perceived effort may not come from a lack of ability but rather from increased critical reflection and expectations for system transparency and pedagogical alignment. Future research should further investigate this counterintuitive finding by exploring how educational background shapes teachers' cognitive processing and attitudes toward AI integration across various teaching contexts.

Thus, it's crucial to address the concerns of MA teachers about AI complexity and reduce barriers to its use. While planning to advance AI literacy among teachers,

stakeholders, and in professional development programs, it is essential to highlight the aspects of AI tools and provide clearer explanations of AI processes for advanced-degree teachers.

According to the hierarchical multiple regression analysis, the sub-scales of AI acceptance had the greatest impact on explaining the variance in AI literacy among teachers. These findings emphasize that teachers' acceptance of AI technology, specifically hedonic motivation, and willingness to use AI, plays a crucial role in determining their AI literacy levels, suggesting that emotional and attitudinal factors may be just as important as technical competence in shaping teachers' overall AI literacy and their capacity to integrate AI tools in educational settings effectively. Therefore, efforts to enhance teachers' AI literacy should prioritize improving their acceptance of AI technology rather than focusing solely on technical training or access to technology. Additionally, since hedonic motivation and willingness to use AI appear essential, training programs could benefit from beginning with engaging, low-stakes applications that allow teachers to explore the enjoyable aspects of AI usage before progressing to more complex implementations. Furthermore, as effort expectancy's association with AI literacy is associated with educational background, support strategies should be tailored differently for teachers with varying educational levels, with a specific focus on reducing perceived barriers to AI use among teachers with master's degrees.

H2 was accepted, affirming the positive association between computational thinking components and AI literacy. Specifically, critical thinking showed the strongest positive correlation with AI literacy, followed by algorithmic thinking, cooperativity, and creativity. This suggests that teachers with stronger computational thinking skills, particularly in critical thinking, demonstrate higher levels of AI literacy. As Celik (2023) confirmed in his research, computational thinking is a significant determinant of AI literacy, which facilitates using, recognizing, and evaluating AI-based technologies. Our research contrasts Celik's (2023) findings on problem-solving, as he deemed it contributory, whereas our study revealed no substantial connection with AI literacy. Nonetheless, this study concurs with his conclusion that stronger computational thinking skills enhance technology usage and digital literacies.

Although computational thinking showed significant positive correlations with AI literacy at the Pearson correlation level, its contribution became negligible once the AI acceptance variables were included in the hierarchical regression. This apparent contradiction suggests that computational thinking may be associated with AI literacy indirectly, through teachers' acceptance, motivation, or willingness to use AI, rather than exerting a direct effect. In other words, teachers who possess stronger computational thinking skills may be more open to adopting AI tools. Future research should test this possibility using mediation or structural equation modeling to examine whether AI acceptance mediates or moderates the relationship between computational thinking and AI literacy.

H3 was accepted, confirming the negative link between AI anxiety and AI literacy, indicating that teachers who experienced higher levels of anxiety about AI technologies showed lower levels of AI literacy. This finding aligns with recent research by Schiavo, et al. (2024), who found similar negative associations between AI anxiety and AI literacy in the general population, as well as with Cengiz and Peker's (2025) findings among

university students. Our results extend these findings to the educational context, specifically among teachers.

The negative correlation between AI anxiety and AI literacy in our study supports the broader understanding of how technological anxieties can impact engagement with and mastery of new technologies (Li & Huang, 2020; Meuter et al., 2003). This relationship is particularly noteworthy given that AI anxiety encompasses unique components beyond general technology anxiety, including concerns about privacy violations, algorithmic bias, job displacement, and the challenge of keeping pace with rapid technological changes (Wang & Wang, 2022).

However, according to the hierarchical multiple regression analysis, AI anxiety did not significantly contribute to the explained variance in teachers' AI literacy. This result may arise from the complex interaction between AI anxiety and various factors associated with AI literacy. Thus, interventions to improve teachers' AI literacy might be more effective when focusing on practical engagement and acceptance rather than solely addressing anxiety-related barriers.

H4 was accepted, confirming the link between the digital divide and AI literacy. This indicates that teachers with improved access to information and communication technologies show higher levels of AI literacy. This finding supports Celik (2023), who noted that better access to information and communication technologies leads to higher AI literacy. Our results further support the established relationship between digital access and technological literacy documented in the digital divide literature (Scheerder et al., 2017; Van Dijk, 2006;).

The positive correlation between digital divide and AI literacy in our study reinforces the broader understanding of how first-level digital divide factors are associated with technological competency (Jung et al., 2001; Segev & Ahituv, 2010). This relationship is particularly relevant in the context of AI technologies, as recent research has highlighted that reliable access to digital infrastructure is crucial for engaging with advanced AI tools (Khowaja et al., 2024).

Again, when referring to the hierarchical multiple regression analysis, results reveal that Digital divide did not significantly contribute to the explained variance of teachers' AI literacy. This finding may be attributed to the fact that, while digital divide correlates with AI literacy, other factors might play a more significant role when examining teachers' AI literacy.

Additional findings

Regarding teachers' educational levels, teachers with master's degrees show higher AI literacy overall, supporting previous findings (Zhao et al., 2022). However, the significant negative relationship between effort expectancy and AI literacy among these teachers is particularly noteworthy and could suggest that teachers with advanced academic education might be more critical or analytical about the effort required to use AI, possibly due to the higher academic standards they hold for themselves or greater awareness of potential implementation difficulties.

As for teaching subjects, it was one of the few background variables that consistently showed a significant association with AI literacy. Science and Technology teachers demonstrated higher AI literacy than their peers in math, language, or the arts. However,

this advantage was primarily observed among male teachers, while female Science and Technology teachers did not display this difference. Although some differences were observed, they may stem from factors related to educational context, working conditions, or training opportunities rather than from inherent gender characteristics. As this is an initial finding, further research is needed to examine these patterns in larger and more diverse samples and to explore the associations with environmental and cultural factors on the relationship between gender and AI literacy.

Regarding AI usage for work, teachers who used AI for work demonstrated significantly higher AI literacy scores compared to those who did not, with this factor emerging as the strongest predictor among the background variables. This finding suggests that teachers engaging with AI tools develop a better understanding through hands-on experience. This finding corresponds with a prior study that shows active engagement with AI technologies enhances AI literacy (Wijaya et al., 2024). Direct interaction with AI applications provides practical learning through trial and error, allowing teachers to observe AI's capabilities and limitations firsthand.

Conclusion

This research investigated the variables that are associated with teachers' AI literacy. Enhancing AI literacy among teachers means confidently utilizing AI tools to refine teaching strategies and preparing students to become responsible AI users in their future careers. By uncovering these variables, this study enables schools and stakeholders to design targeted, efficient, and professional development, rather than using a one-size-fits-all approach that may waste resources and frustrate teachers. This targeted approach may suggest that professional development programs need to be differentiated based on teachers' individual characteristics, experience levels, and attitudes toward AI technology, which ensures that support systems help teachers gain the specific AI skills they need, making them more likely to successfully integrate AI tools in ways that benefit both their teaching practice and their students' learning outcomes.

Theoretical and practical implications

The current study presents some theoretical implications. First, the findings enrich AI literacy theory (Ng et al., 2021; Wang et al., 2022) by empirically validating that literacy does not develop solely from cognitive exposure or technical training but also from attitude and acceptance that promote sustained engagement and reflective use. This emphasizes the role of self-determined motivation and intrinsic curiosity as catalysts for literacy acquisition, aligning with emerging frameworks that view AI literacy as dynamic and experience-driven (Celik, 2023; Chiu et al., 2024).

Secondly, this study establishes differences between AI acceptance components (Goursey et al., 2019) and AI literacy among teachers, with hedonic motivation and willingness to use AI, emerging as the strongest predictors. This extends existing technology acceptance theories by specifically mapping the association between different acceptance variables and AI literacy in educational contexts.

Thirdly, the findings challenge existing theoretical frameworks regarding the role of computational thinking in AI literacy (Celik, 2023), positioning computational thinking as a direct determinant of AI literacy. This research uncovers a more

intricate connection; although the components of computational thinking initially show a positive correlation with AI literacy, these links were weakened when considering additional variables. This highlights the need for more nuanced theoretical models to better capture the connection between computational thinking and AI literacy.

Furthermore, the study advances theoretical understanding of AI anxiety by demonstrating that while it correlates with AI literacy, its explanatory power diminishes when controlling for other variables. This suggests a need to revise theoretical models that place heavy emphasis on anxiety as a barrier to technology adoption.

Finally, by showing that the digital divide's effect diminishes once acceptance and motivational variables are considered, the study offers new theoretical insight into digital inequality theory (Van Dijk, 2020). It suggests that the contemporary digital divide is no longer primarily structural (access-based) but psychological and affective, emerging from disparities in confidence, curiosity, and perceived enjoyment of AI use.

Collectively, these contributions point toward an integrative theoretical model of AI Literacy Engagement, where cognitive, affective, and contextual variables jointly determine teachers' readiness to engage with AI in educational practice.

Regarding practical implementation, stakeholders, professional development programs, and individual teachers could gain insights from the study's findings. First, programs designed to enhance teachers' AI literacy should move beyond traditional, technical training toward experiential and motivational approaches that emphasize enjoyment, curiosity, and reflective use of AI tools. Beginning with low-stakes, engaging applications can increase hedonic motivation and willingness to use AI, thereby fostering both confidence and competence.

Secondly, professional development programs should be expanded and made more inclusive to accommodate the diverse needs of teachers across a wide range of backgrounds, disciplines, and perceived effort levels. For instance, while teachers with advanced degrees may benefit from training that emphasizes conceptual clarity and ethical reflection, teachers without advanced education might need different types of support, such as practical classroom management strategies. Additionally, offering a variety of targeted workshops, covering non-technical subjects and other areas, can help reduce perceived barriers, foster greater collaboration among teachers, and promote equitable participation and professional growth across all disciplines. This comprehensive approach ensures that professional development is not only tailored but also accessible to all educators, ultimately leading to a more effective and cohesive educational environment.

Thirdly, policy-level interventions are needed to integrate AI literacy within teacher education curricula and national digital strategies. Institutional support, such as dedicated time, infrastructure, and access to digital resources, can mitigate effort expectancy and strengthen teachers' engagement with AI.

Finally, teacher education institutions should adopt design principles for AI literacy development that combine hands-on experimentation, collaborative reflection, and critical dialogue about ethical and social dimensions of AI use, aligning with frameworks such as TPACK and UNESCO's (2024) AI Literacy model.

Research limitations and future research

The study presents several important limitations that should be taken into account when interpreting its findings. The research provides a snapshot of AI literacy and its predictors, preventing the observation of how these relationships evolve over time or how teachers' AI literacy develops through different stages of technology adoption. To address this limitation, future research could conduct longitudinal studies monitoring teachers' AI literacy development.

Moreover, the study's reliance on self-reported data regarding AI literacy and usage poses a limitation, as it may introduce bias, where participants may exaggerate their AI knowledge or usage to appear more technologically proficient. To address self-reporting biases, future research could incorporate direct observational data on teachers' AI tool usage in authentic classroom settings, thereby combining both self-reported and behavioral measures.

Furthermore, because the sample was drawn from a consumer research panel of Israeli teachers, it may overrepresent participants who are more technologically inclined or comfortable with digital platforms, potentially introducing a bias toward higher AI literacy levels within the sample.

The current study concentrates on a specific set of variables linked to previous literacies and studies regarding AI literacy. However, future research could explore more personality and cognitive factors to create a more comprehensive model of how teachers' individual differences relate to their AI literacy. Additionally, future research could examine other countries and cultures.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40561-026-00433-5.

Additional file1 (PDF 162 KB)

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Mor Deshen: Conceptualization, Data curation, Methodology, Writing – original draft. Ruben Harari: Conceptualization, Formal analysis, Methodology, Writing – review and editing. Noa Aharony: Conceptualization, Data curation, Methodology, Writing – review and editing.

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The data generated during the current study could be available upon a reasonable request.

Declarations

Competing interests
The authors report that they have no competing interests to declare.

Ethical approval
This study was reviewed and approved by the Ethics Committee of the Academic College of Ramat Gan (Approval No. 1018). All procedures involving human participants were conducted in accordance with the ethical standards of the institutional research committee and the 1964 Helsinki Declaration and its subsequent amendments.

Informed consent
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